PHYS 210 Electricity and Magnetism

Celebration of Knowledge!

Practice Midterm G - CK #2 preparation

Name: _____ Date: _____

Time: _____ Marks: _____ Practice paper - not an official exam

Choose 3 of the following 4 problems. Complete every part of each chosen problem.

Write all rough calculations and final answers on separate paper. Show your working.

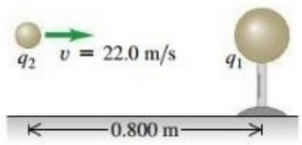
Original source numbering is retained inside each question. Answer the entire cropped question.

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1. [____] Homework 5 - Question 23.5

23.5 •• A small metal sphere, carrying a net charge of $q_1 = -2.80\ \mu C$, is held in a stationary position by insulating supports. A second small metal sphere, with a net charge of $q_2 = -7.80\ \mu C$ and mass 1.50 g, is projected toward q_1. When the two spheres are 0.800 m apart, q_2, is moving toward q_1 with speed 22.0 m/s (**Fig. E23.5**). Assume that the two spheres can be treated as point charges. You can ignore the force of gravity. (a) What is the speed of q_2 when the spheres are 0.400 m apart? (b) How close does q_2 get to q_1?

Figure **E23.5**



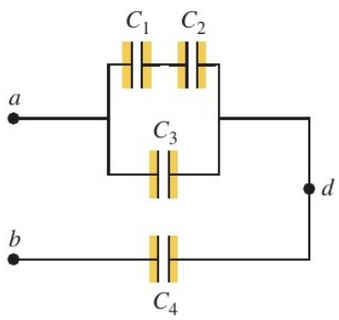
2. [____] Textbook 15e - Question 22.25

22.25 • A conductor with an inner cavity, like that shown in Fig. 22.23c, carries a total charge of +5.00 nC. The charge within the cavity, insulated from the conductor, is −6.00 nC. How much charge is on (a) the inner surface of the conductor and (b) the outer surface of the conductor?

3. [____] Textbook 15e - Question 24.52

24.52 •• In Fig. E24.17, $C_1 = 6.00\ \mu F$, $C_2 = 3.00\ \mu F$, $C_3 = 4.00\ \mu F$, and $C_4 = 8.00\ \mu F$. The capacitor network is connected to an applied potential difference V_{ab}. After the charges on the capacitors have reached their final values, the voltage across C_3 is 40.0 V. What are (a) the voltages across C_1 and C_2, (b) the voltage across C_4, and (c) the voltage V_{ab} applied to the network?

Figure **E24.17**



4. [____] Textbook 15e - Question 24.44

24.44 • A parallel-plate capacitor has plates with area 0.0225 m^2 separated by 1.00 mm of Teflon. (a) Calculate the charge on the plates when they are charged to a potential difference of 12.0 V. (b) Use Gauss's law [Eq. (24.23)] to calculate the electric field inside the Teflon. (c) Use Gauss's law to calculate the electric field if the voltage source is disconnected and the Teflon is removed.